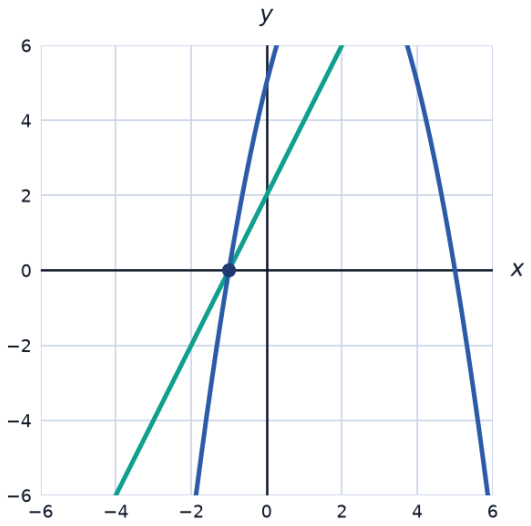
Indices, Surds & Quadratics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Laws of Indices & Surd Arithmetic

The diagram shows the graphs of the linear equation $y = 2x + c$ and the quadratic equation $y = -x^2 + 4x + 5$. The two graphs intersect at the points where $x = -1$ and $x = k$. Find the value of c and the value of k.



2 [Foundational] • Target: Completing the Square & Turning Points

The points A, B, and C have coordinates A(2, 5), B(k, 1), and C(8, -3). If A, B, and C are collinear, find the value of k.

3 [Foundational] • Target: The Discriminant & Nature of Roots

The graph of $y = f(x)$ is a curve passing through the points A(1, 2), B(4, 2), and C(2, 5). This graph is then subjected to a sequence of transformations. First, it is reflected in the line $y = x$. Let the image of the points A, B, and C under this reflection be A', B', and C' respectively. Secondly, the graph of the reflected curve is translated by the vector $(-3; 0)$. Let the final image of the points A', B', and C' after this translation be A'', B'', and C''. Given that the original curve is a parabola with its axis of symmetry parallel to the y-axis, and passes through A, B, and C, find the coordinates of C''.

4 [Foundational] • Target: Disguised Quadratics & Exam-Style Modeling

The diagram shows the graph of the function $y = f(x)$. The coordinates of the points A, B, and C are given. The graph is transformed by a translation of 3 units to the right and 2 units down. What are the new coordinates of point A?

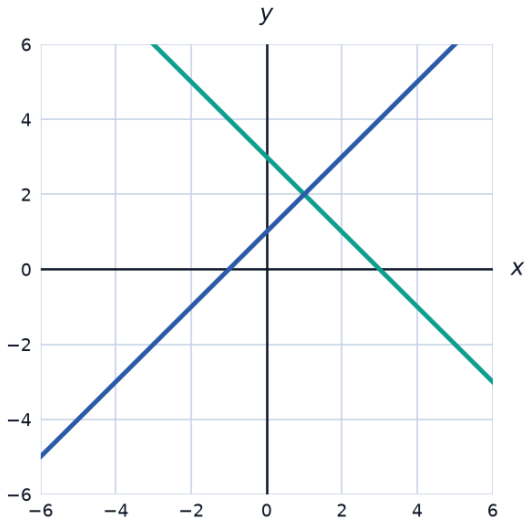
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5 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Given the quadratic equation $x^2 - 5x + 6 = 0$, find the sum of the roots.

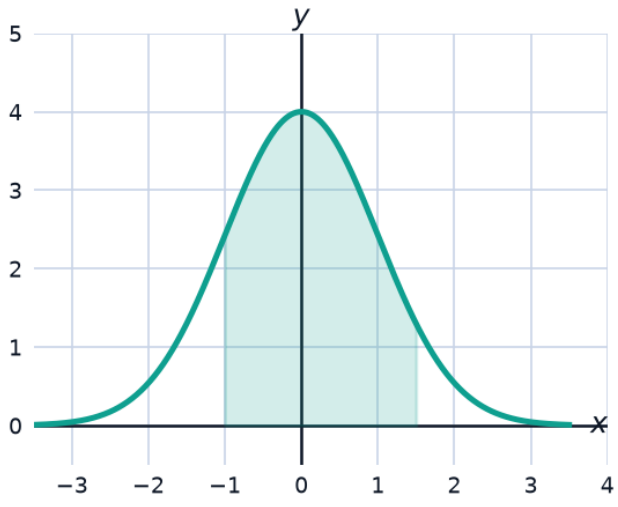
6 [Foundational] • Target: Completing the Square & Turning Points

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



7 [Foundational] • Target: The Discriminant & Nature of Roots

The weights of bags of sugar are normally distributed with a mean of 1000 g and a standard deviation of 5 g. Let X be the weight of a randomly selected bag of sugar. Calculate the probability that a bag of sugar weighs between 990 g and 1010 g.

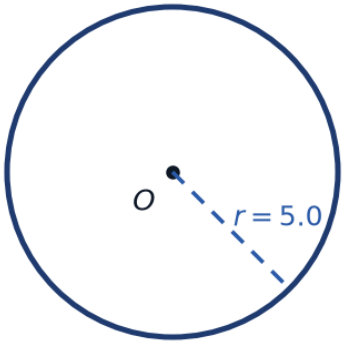


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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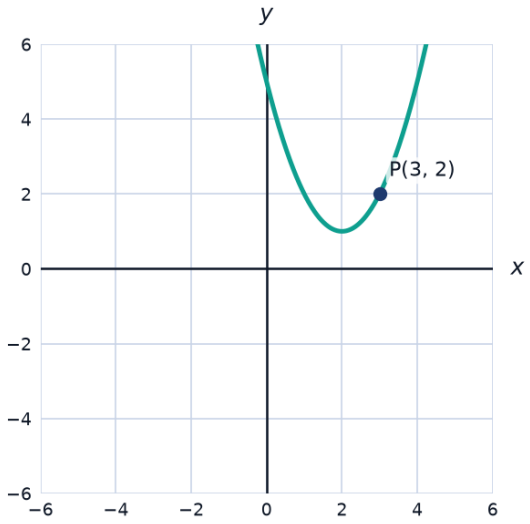
8 **[Intermediate]** • **Target: Disguised Quadratics & Exam-Style Modeling**

The diagram shows a sector of a circle with centre O. The arc length of the sector is given by 6π cm and the area of the sector is 27π cm^2. The radius of the circle is r cm and the angle of the sector is θ radians. Find the value of r and θ.



9 **[Intermediate]** • **Target: Laws of Indices & Surd Arithmetic**

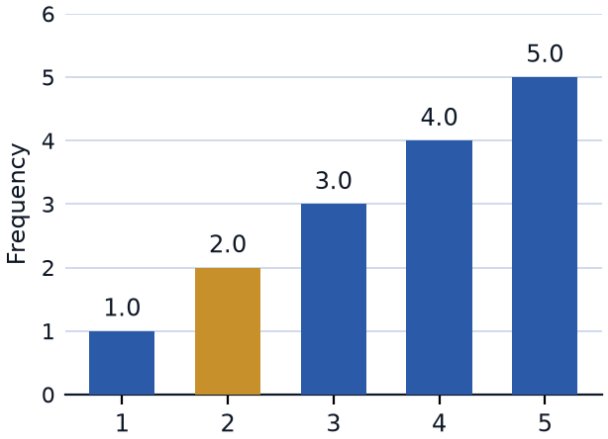
The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point P.



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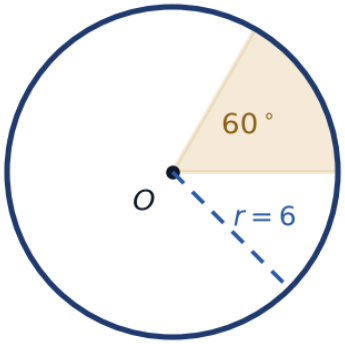
10 [Intermediate] • Target: Completing the Square & Turning Points

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.



11 [Intermediate] • Target: The Discriminant & Nature of Roots

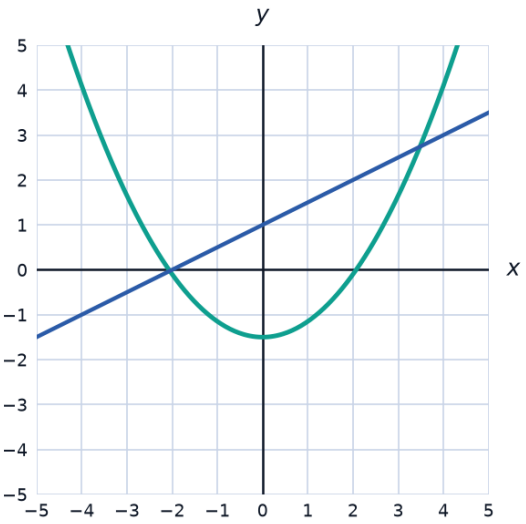
Solve $\sin(x) = \frac{1}{2}$ for $0^\circ \leq x \leq 360^\circ$.



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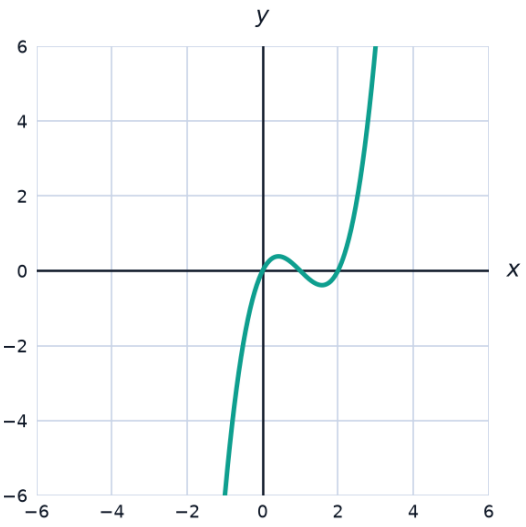
12 **[Intermediate]** • **Target: Disguised Quadratics & Exam-Style Modeling**

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6*cmandtheradiusisUSD*3 cm, find the angle θ in radians.



13 **[Intermediate]** • **Target: Laws of Indices & Surd Arithmetic**

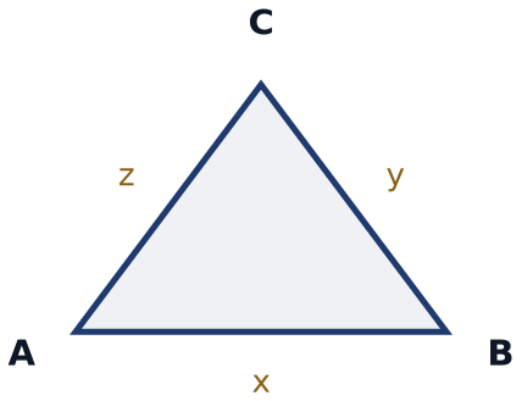
A curve is given by the equation $y = x^3 - 3x^2 + 2x$. Find the gradient of the curve at $x = 2$.



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14 [Intermediate] • Target: Completing the Square & Turning Points

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 6.01 Differentiation & Integration

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

16 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 7.01 Statistics & Probability

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

17 [Challenging] • Target: Laws of Indices & Surd Arithmetic • Cross-Lesson: 8.01 Functions: Composite & Inverse

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

18 [Challenging] • Target: Completing the Square & Turning Points • Cross-Lesson: 9.01 Circular Measure

A particle of mass 8 kg is moving under the action of a net force $\mathbf{F}$ N. The acceleration of the particle, $\mathbf{a}$ m/s^2, is given by the vector $\mathbf{a} = (2t - 4; t^2 - 9)$ for $t \geq 0$, where t is the time in seconds. The particle is initially at rest at the origin.

19 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 10.01 Series & Binomial Theorem

A sector of a circle with radius r cm has a perimeter of P cm. The area of the sector is A cm^2. Prove that $P^2 \geq 16A$.

20 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Simplify $\left(\frac{8x^6}{27y^3}\right)^{1/3}$ completely.

a. $\frac{2x^2}{3y}$

b. $\frac{4x^2}{9y}$

c. $\frac{2x^3}{3y}$

d. $\frac{8x^2}{27y}$

22 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Simplify $\sqrt{48} - \sqrt{12} + \sqrt{75}$ into the form $k\sqrt{3}$.

a. $7\sqrt{3}$

b. $5\sqrt{3}$

c. $9\sqrt{3}$

d. $6\sqrt{3}$

23 [Foundational] • Target: Laws of Indices & Surd Arithmetic

Rationalize the denominator of $\frac{6}{3-\sqrt{3}}$ and write in simplest form.

a. $3+\sqrt{3}$

b. $2+\sqrt{3}$

c. $3-\sqrt{3}$

d. $6+2\sqrt{3}$

24 [Foundational] • Target: Completing the Square & Turning Points

Express $2x^2 - 12x + 11$ in the form $a(x-h)^2 + k$.

a. $2(x-3)^2 - 7$

b. $2(x-3)^2 + 11$

c. $2(x-6)^2 - 25$

d. $2(x-3)^2 - 11$

25 [Foundational] • Target: Completing the Square & Turning Points

Find the minimum value of $f(x) = 3x^2 + 18x + 20$ and the value of x at which it occurs.

a. Minimum is -7 at $x = -3$

b. Minimum is 20 at $x = 0$

c. Minimum is -7 at $x = 3$

d. Minimum is 7 at $x = -3$

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26 [Foundational] • Target: Completing the Square & Turning Points

The curve $y = -(x + 4)^2 + 15$ has a stationary point. State its nature and coordinates.

a. Maximum at $(-4, 15)$

b. Minimum at $(-4, 15)$

c. Maximum at $(4, 15)$

d. Minimum at $(4, -15)$

27 [Foundational] • Target: The Discriminant & Nature of Roots

Find the set of values of k for which the equation $x^2 + kx + (k + 3) = 0$ has two distinct real roots.

a. $k < -2$ or $k > 6$

b. $-2 < k < 6$

c. $k \leq -2$ or $k \geq 6$

d. $k > 6$

28 [Foundational] • Target: The Discriminant & Nature of Roots

The line $y = 2x + c$ is tangent to the curve $y = x^2 - 4x + 7$. Find the exact value of c.

a. -2

b. 2

c. 7

d. -6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Discriminant & Nature of Roots

Determine the condition on p such that the quadratic $px^2 + 4x + p = 0$ has no real roots, where $p > 0$.

a. $p > 2$

b. $0 < p < 2$

c. $p > 4$

d. $p < -2$

30 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

Solve the disguised quadratic equation $x^4 - 5x^2 - 36 = 0$ for all real solutions.

a. $x = \pm 3$

b. $x = \pm 2$

c. $x = \pm 3, \pm 2i$

d. $x = 9$

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31 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

Solve the equation $2^{2x} - 10(2^x) + 16 = 0$.

a. $x = 1$ or $x = 3$

b. $x = 2$ or $x = 8$

c. $x = 0$ or $x = 4$

d. $x = -1$ or $x = 3$

32 [Intermediate] • Target: Disguised Quadratics & Exam-Style Modeling

A rectangular enclosure is formed using 80 meters of fencing against a straight stone wall. What is the maximum possible area of the enclosure in square meters?

a. $800\ \text{m}^2$

b. $400\ \text{m}^2$

c. $1600\ \text{m}^2$

d. $600\ \text{m}^2$

33 [Intermediate] • Target: Laws of Indices & Surd Arithmetic

Find the discriminant of $3x^2 - 5x - 10 = 0$ and state the number of real roots.

a. $\Delta = -145$: two distinct real roots

b. $\Delta = 145$: one repeated real root

c. $\Delta = 153$: no real roots

d. $\Delta = 145$: two distinct real roots

34 [Intermediate] • Target: Completing the Square & Turning Points

Find the discriminant of $x^2 - 4x - 7 = 0$ and state the number of real roots.

a. $\Delta = 52$: no real roots

b. $\Delta = 44$: one repeated real root

c. $\Delta = 44$: two distinct real roots

d. $\Delta = -44$: two distinct real roots

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35 **[Intermediate]** • **Target: The Discriminant & Nature of Roots**

Find the discriminant of $x^2 - 6x - 2 = 0$ and state the number of real roots.

a. $\Delta = -44$: two distinct real roots

b. $\Delta = 44$: two distinct real roots

c. $\Delta = 52$: no real roots

d. $\Delta = 44$: one repeated real root

36 **[Intermediate]** • **Target: Disguised Quadratics & Exam-Style Modeling**

Find the discriminant of $x^2 - 4x - 6 = 0$ and state the number of real roots.

a. $\Delta = 40$: one repeated real root

b. $\Delta = -40$: two distinct real roots

c. $\Delta = 48$: no real roots

d. $\Delta = 40$: two distinct real roots

37 **[Intermediate]** • **Target: Laws of Indices & Surd Arithmetic**

Find the discriminant of $4x^2 - 5x - 2 = 0$ and state the number of real roots.

a. $\Delta = -57$: two distinct real roots

b. $\Delta = 57$: two distinct real roots

c. $\Delta = 57$: one repeated real root

d. $\Delta = 65$: no real roots

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Completing the Square & Turning Points • Cross-Lesson: 9.01 Circular Measure

Find the discriminant of $5x^2 - 6x - 3 = 0$ and state the number of real roots.

a. $\Delta = 104$: no real roots

b. $\Delta = 96$: two distinct real roots

c. $\Delta = -96$: two distinct real roots

d. $\Delta = 96$: one repeated real root

39 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 10.01 Series & Binomial Theorem

Find the discriminant of $5x^2 + 8x - 5 = 0$ and state the number of real roots.

a. $\Delta = 164$: one repeated real root

b. $\Delta = 172$: no real roots

c. $\Delta = -164$: two distinct real roots

d. $\Delta = 164$: two distinct real roots

40 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 11.01 Kinematics

Find the discriminant of $x^2 + x - 3 = 0$ and state the number of real roots.

a. $\Delta = 21$: no real roots

b. $\Delta = 13$: one repeated real root

c. $\Delta = -13$: two distinct real roots

d. $\Delta = 13$: two distinct real roots

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41 [Challenging] • Target: Laws of Indices & Surd Arithmetic • Cross-Lesson: 12.01 Forces & Equilibrium

Find the discriminant of $3x^2 - 6x + 1 = 0$ and state the number of real roots.

a. $\Delta = 32$: no real roots

b. $\Delta = -24$: two distinct real roots

c. $\Delta = 24$: two distinct real roots

d. $\Delta = 24$: one repeated real root

42 [Challenging] • Target: Completing the Square & Turning Points • Cross-Lesson: 13.01 Newton's Laws

Find the discriminant of $x^2 - 7x + 6 = 0$ and state the number of real roots.

a. $\Delta = 33$: no real roots

b. $\Delta = 25$: one repeated real root

c. $\Delta = -25$: two distinct real roots

d. $\Delta = 25$: two distinct real roots

43 [Challenging] • Target: The Discriminant & Nature of Roots • Cross-Lesson: 14.01 Energy, Work & Power

Find the discriminant of $2x^2 + 7x + 6 = 0$ and state the number of real roots.

a. $\Delta = 1$: two distinct real roots

b. $\Delta = 9$: no real roots

c. $\Delta = -1$: two distinct real roots

d. $\Delta = 1$: one repeated real root

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44 [Challenging] • Target: Disguised Quadratics & Exam-Style Modeling • Cross-Lesson: 15.01 Representation of Data

Find the discriminant of $x^2 + 2x + 1 = 0$ and state the number of real roots.

a. $\Delta = 0$: one repeated real root

b. $\Delta = 8$: no real roots

c. $\Delta = 16$: one repeated real root

d. $\Delta = 0$: two distinct real roots

45 [Challenging] • Target: Laws of Indices & Surd Arithmetic • Cross-Lesson: 16.01 Permutations & Combinations

Find the discriminant of $5x^2 + 10x + 11 = 0$ and state the number of real roots.

a. $\Delta = -120$: two distinct real roots

b. $\Delta = 120$: no real roots

c. $\Delta = -120$: no real roots

d. $\Delta = -112$: one repeated real root

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